UNIVERSITÀ DEGLI STUDI FIRENZE
DINFO
Dipartimento di Ingegneria dell'Informazione

Laboratorio di Elettronica Digitale

Microcontrollers

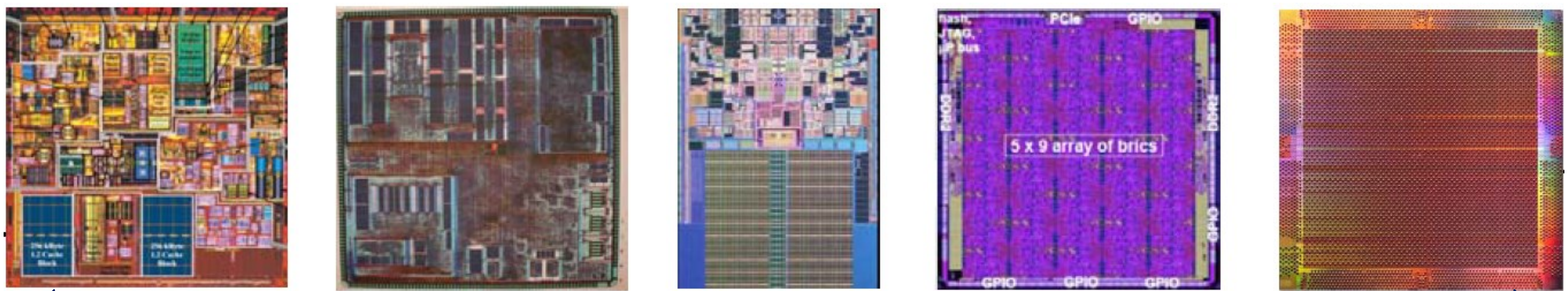
Enrico Boni

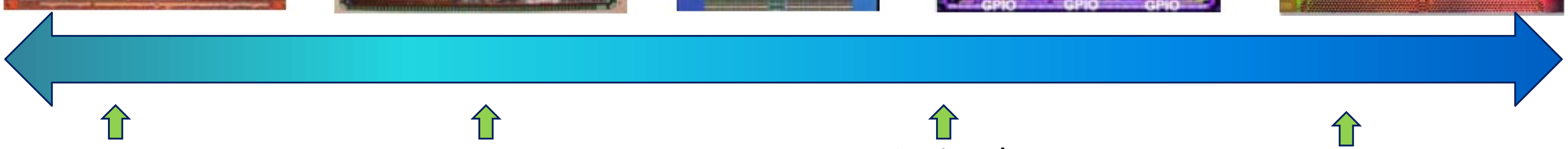
Programmable digital devices

- Microcontrollers (AVR, PIC)
- Microprocessors (ARM, AMD, INTEL)
- Digital signal processors (DSP) (TMS320C6X)
- Graphic Processing Unit (GPU) (NVIDIA Kepler)
- Massively Parallel Processor Arrays (MPPA) (Ambric AM2045)
- Complex Programmable Logic Devices (CPLD) (MAX3000)
- Field Programmable Gate Array (FPGA) (EPS4S, EP5GX)

Programmable digital devices

CPU DSP Multicores Arrays FPGA



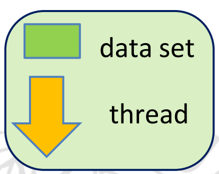


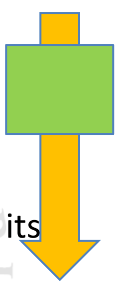
Single core

Multicores Coarse-Grained CPUs and DSPs

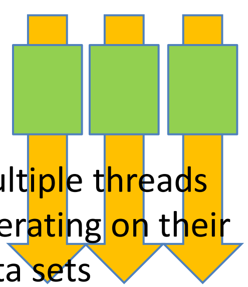
Coarse-Grained Massively Parallel Processor Arrays

Fine-Grained Massively Parallel Arrays

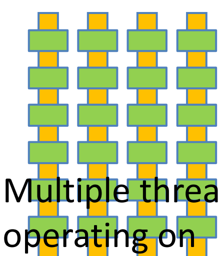




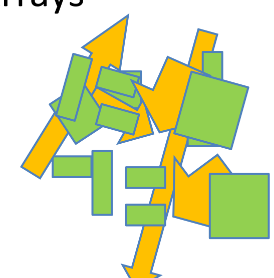
Single thread operating on its data set



Multiple threads operating on their data sets



Multiple threads operating on multiple data sets



'unrelated' threads and data sets

Definitions

- Microprocessor:
  - Single chip processing unit (e.g. 8086)
  - Needs external circuits to work:
    - RAM
    - DMA Controller (Intel 8237)
    - UART (8250)
    - Interrupt controller (Intel 8259)

Definitions

- Microcontroller:
  - Single chip processing unit
    - Usually simpler than a Microprocessor
  - Doesn’t need external circuits to work:
    - Embedded RAM, FLASH, DMA,PIC
    - Rich set of embedded peripherals

MC Requirements

- Fast and predictable code execution
  - RISC core
  - CISC core with small instruction set
  - Harvard vs Von Neumann memory architecture

MC Requirements

- Reduced footprint:
  - Small dimensions, integrated system on chip
  - Power-wise electronics (down to few uW)
    - Battery operated devices
    - Miniaturized power supply circuits

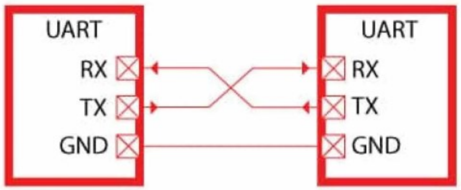
MC Requirements

- Easy interface with external world:
  - GPIO, bit addressable
  - Communication peripherals
  - Mixed signal peripherals (ADC/DAC)
  - Numerical Control interfaces (PWM)
  - Fast reaction to external events (HW IRQ)

MC Requirements

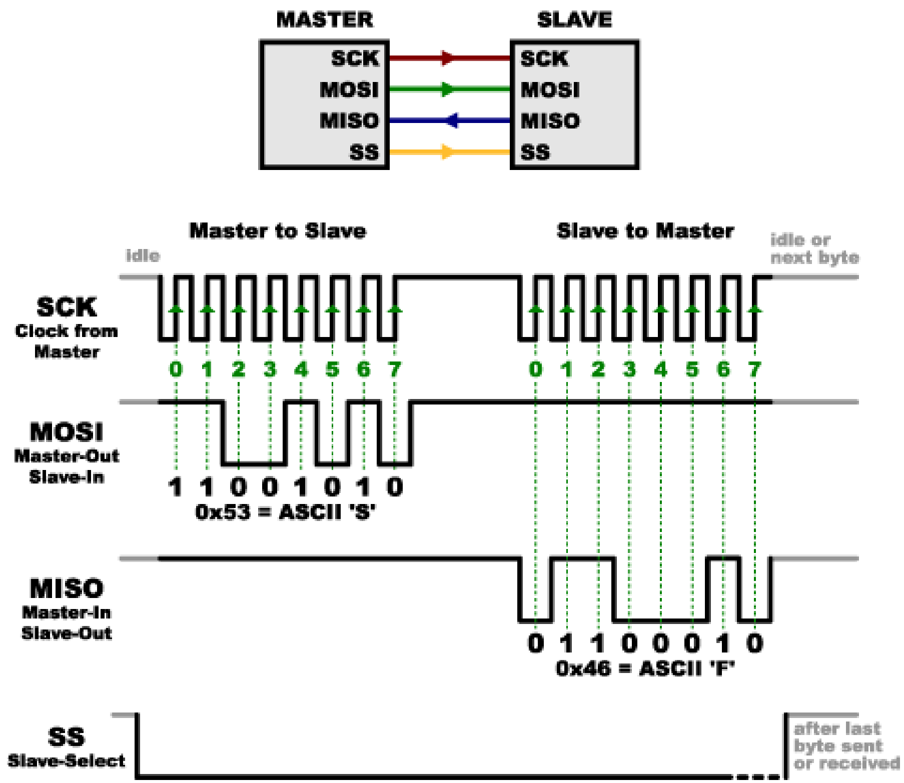
- Communication peripherals:
  - UART, SPI, I2C
  - CAN
  - USB, Ethernet

UART: (Universal Asynchronous Receiver-Transmitter)



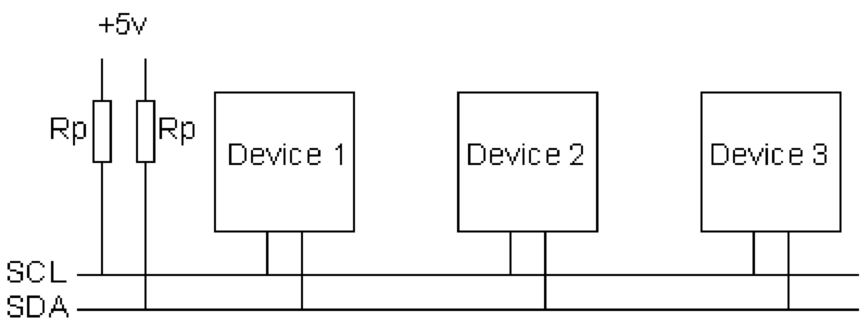
- Asynchronous: Master and slave need to have a ‘common’ timing source.

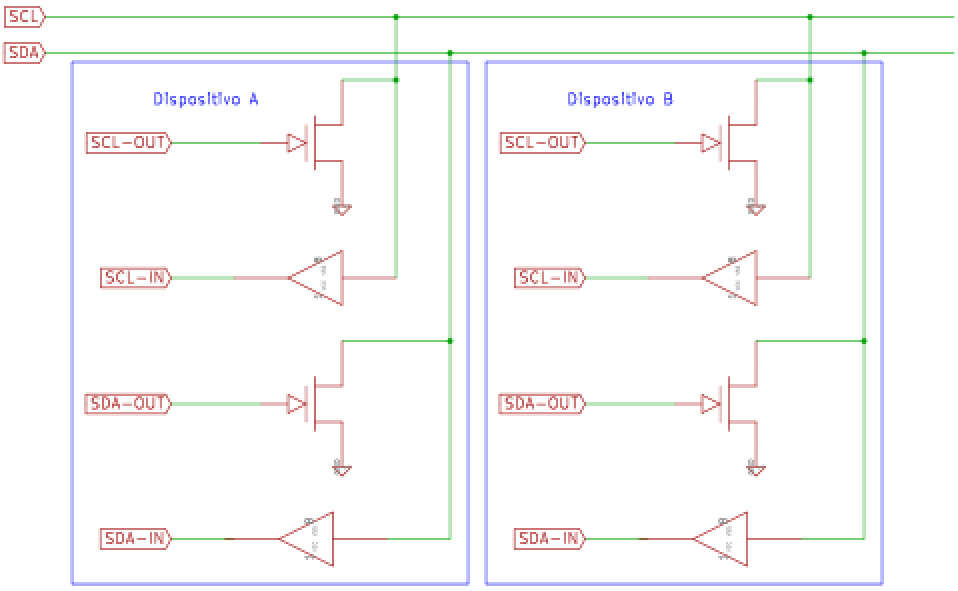
SPI:
(Serial Peripheral Interface)



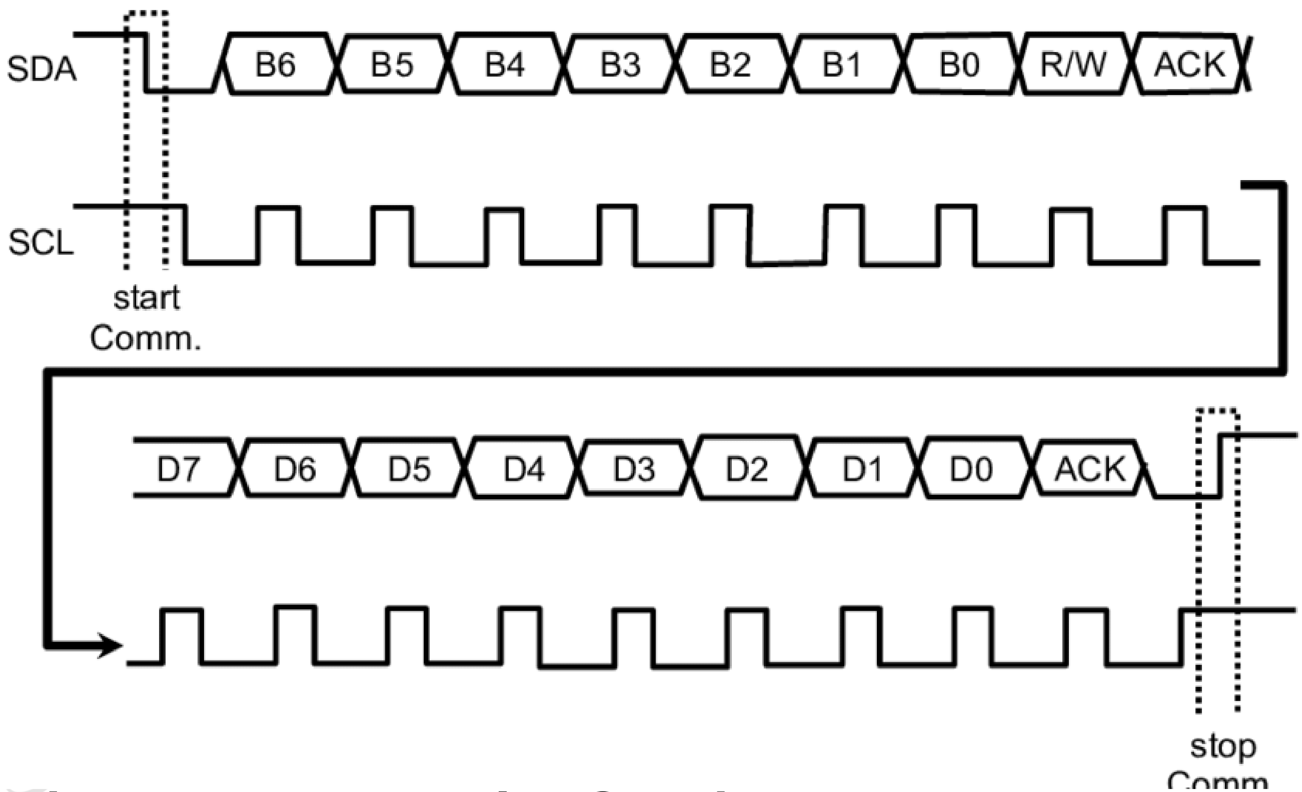
- Synchronous, but more wires.

I2C:
(Inter-Integrated Circuit)





I2C:
(Inter-Integrated Circuit)



- Synchronous, only 2 wires.

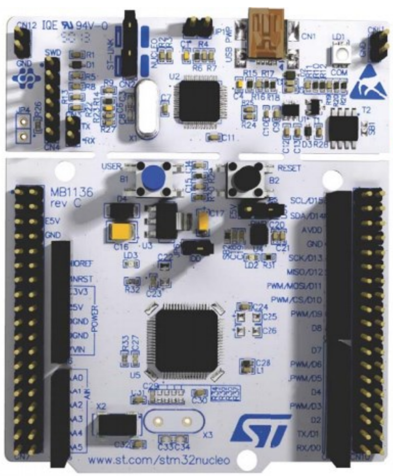
I2C:
(Inter-Integrated Circuit)

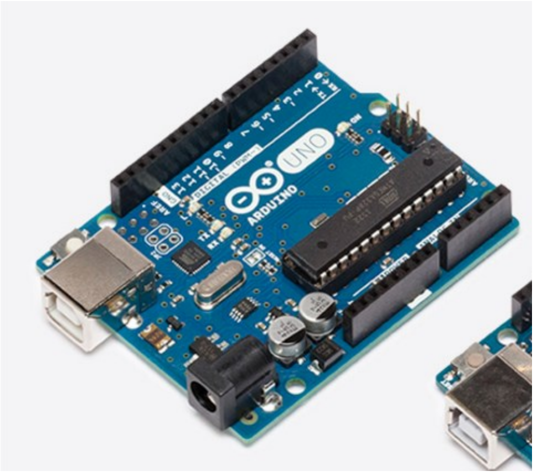
- Typical transactions:
  - Writing a command
  - Writing data to a memory
  - Reading data (from a memory/sensor)

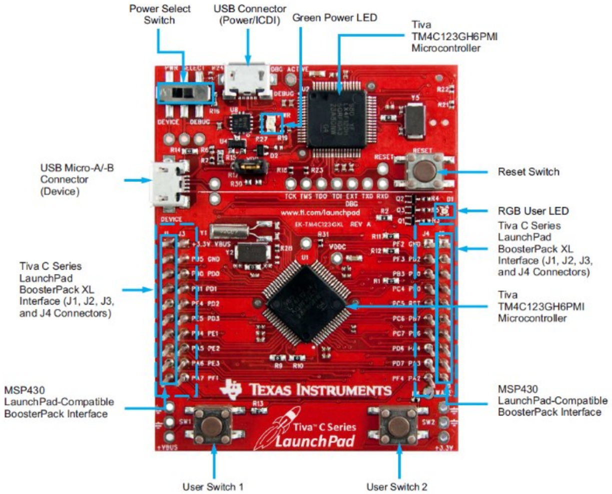
I2C:
(Inter-Integrated Circuit)

- Typical pull-up resistors:
  - 100kbit/s → 10k Ohm
  - 400kbit/s → 2.2k Ohm
  - 1Mbit/s → 1k Ohm

Some uC boards

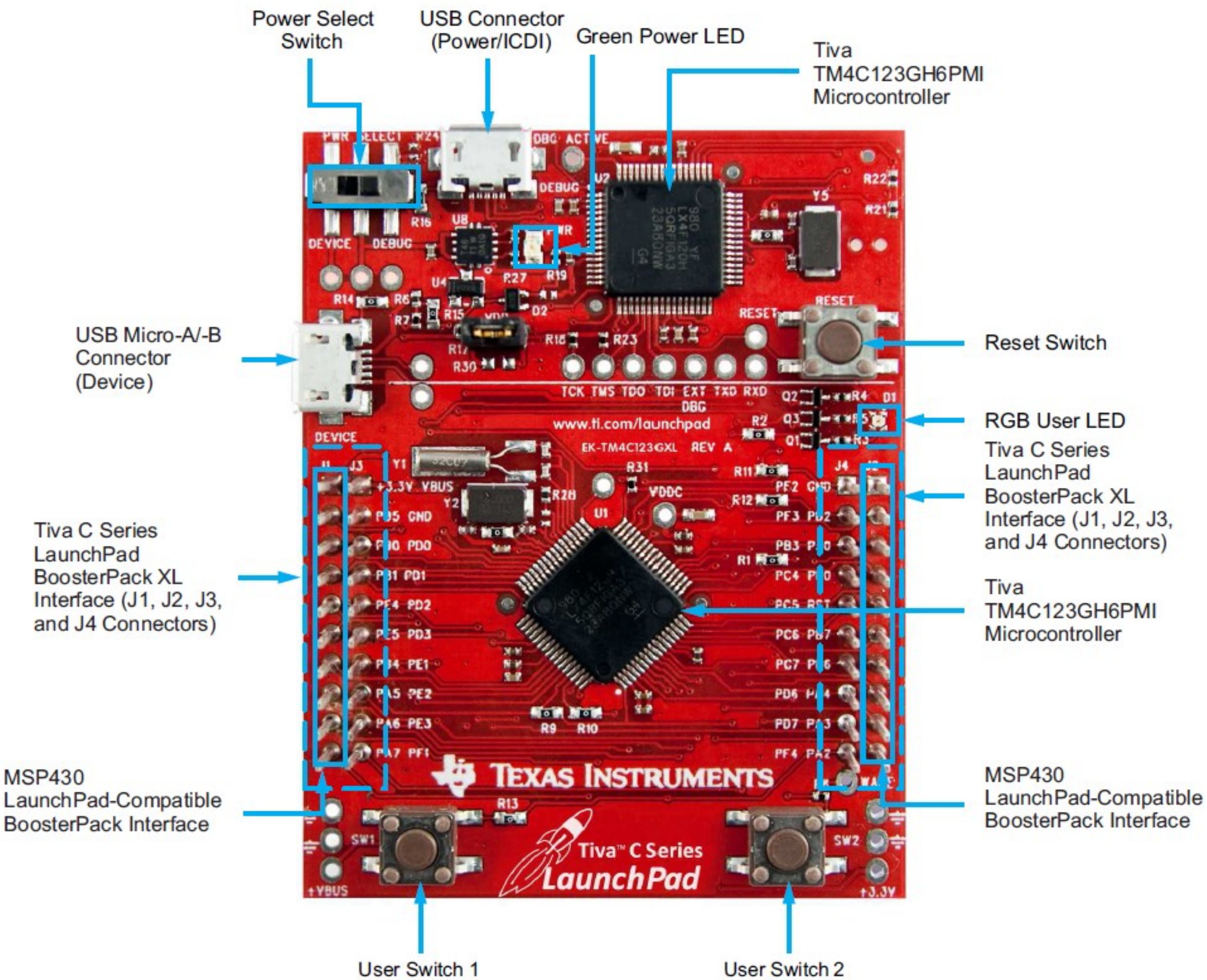




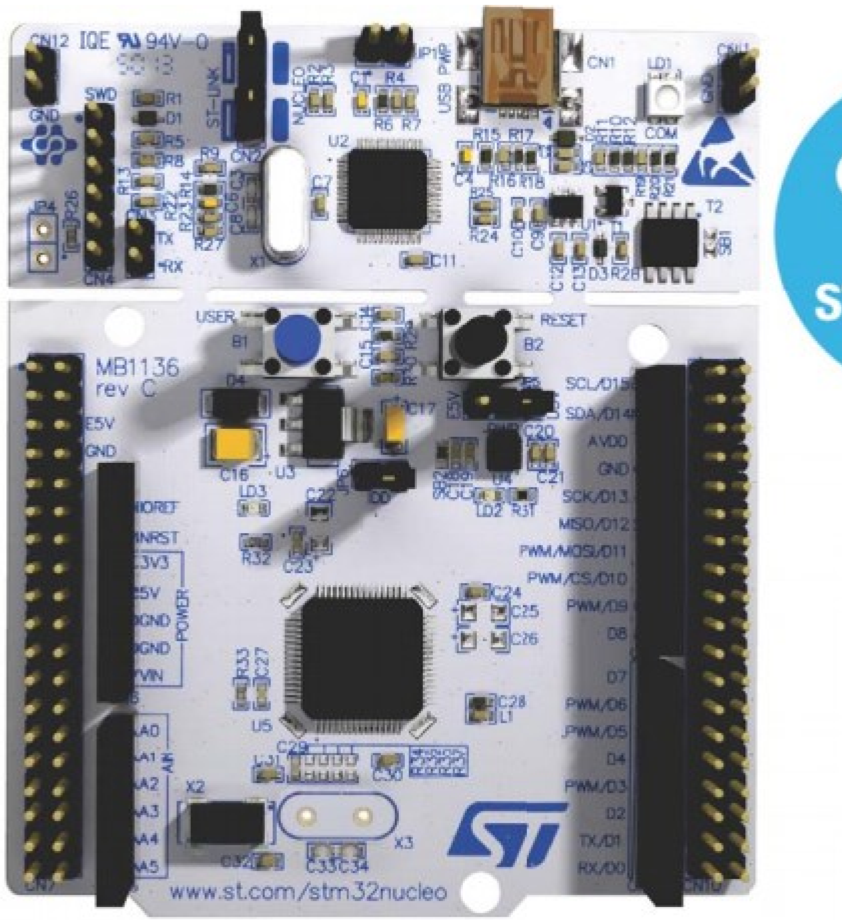


Arduino Board:

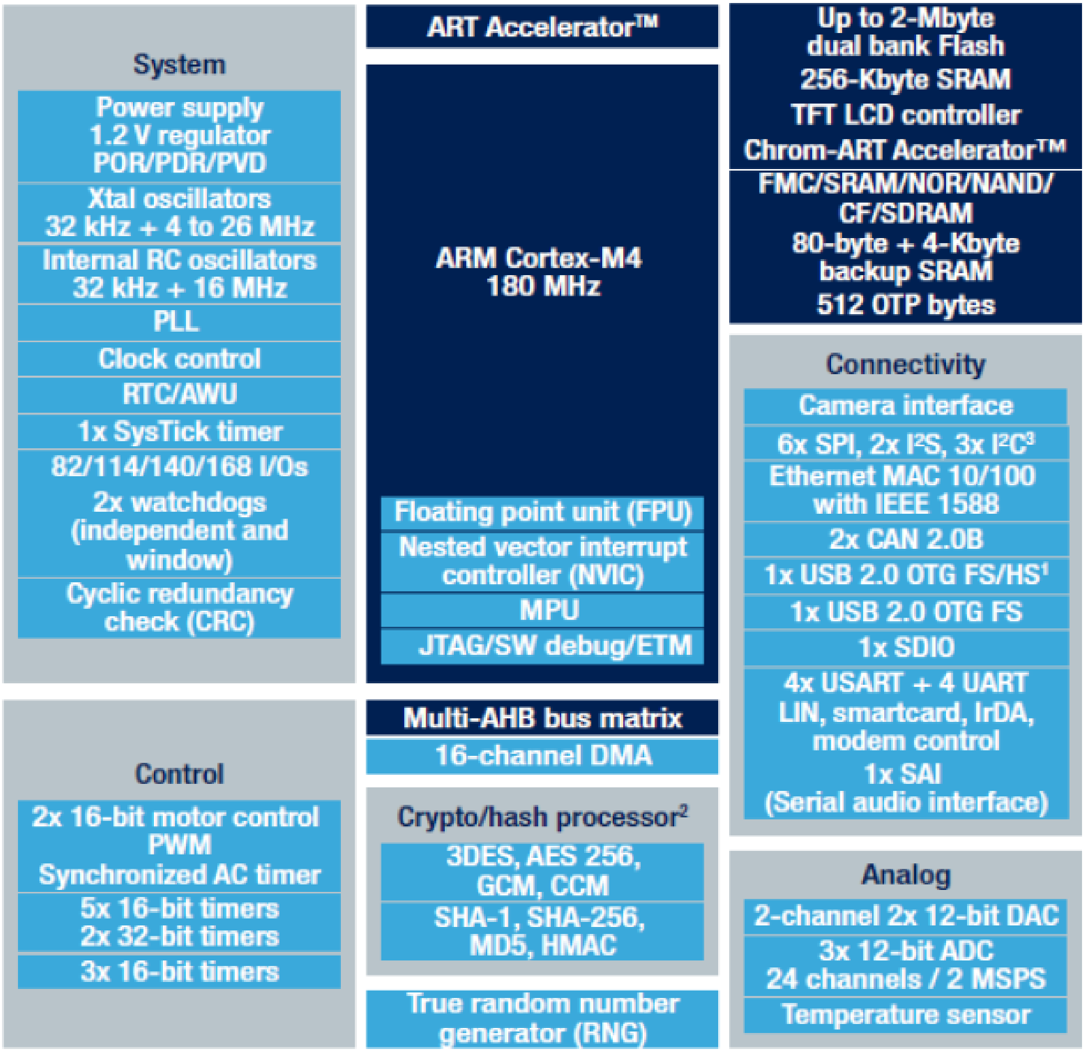
Tiva-c board:


Power Select Switch
USB Connector (Power/ICDI)
Green Power LED
Tiva TM4C123GH6PMI Microcontroller
USB Micro-A/-B Connector (Device)
Reset Switch
RGB User LED
Tiva C Series LaunchPad BoosterPack XL Interface (J1, J2, J3, and J4 Connectors)
Tiva C Series LaunchPad BoosterPack XL Interface (J1, J2, J3, and J4 Connectors)
Tiva TM4C123GH6PMI Microcontroller
MSP430 LaunchPad-Compatible BoosterPack Interface
MSP430 LaunchPad-Compatible BoosterPack Interface
User Switch 1
User Switch 2
TEXAS INSTRUMENTS
Tiva™ C Series LaunchPad
www.ti.com/launchpad
EK-TM4C123GXL REV A

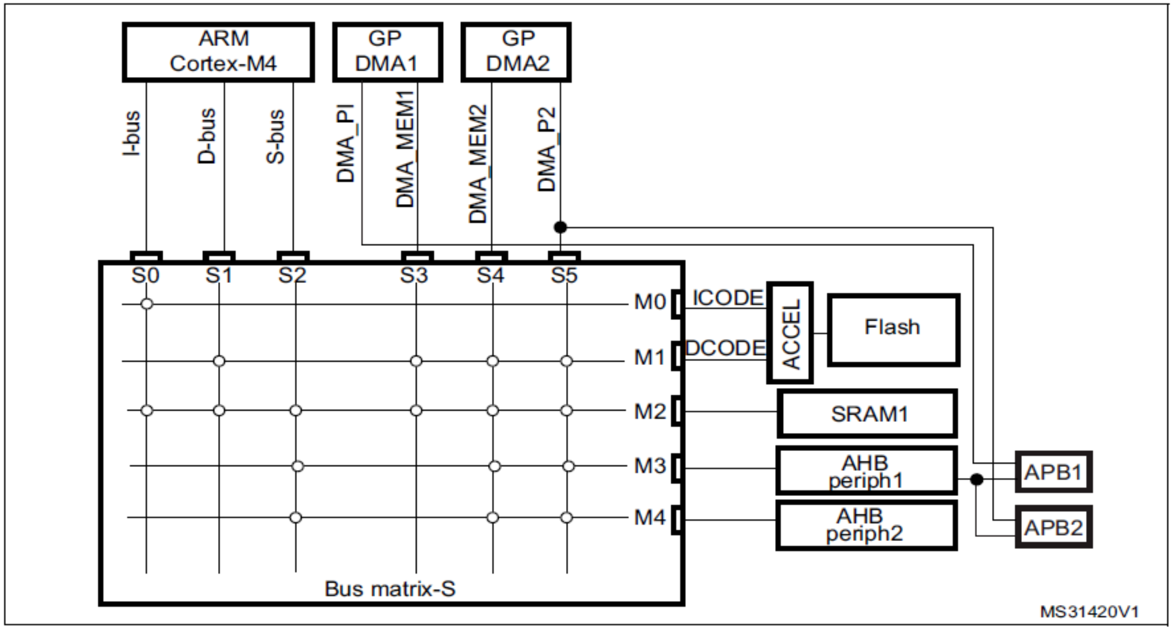
STM32 Nucleo:



STM32 Cortex-M4 (F4)



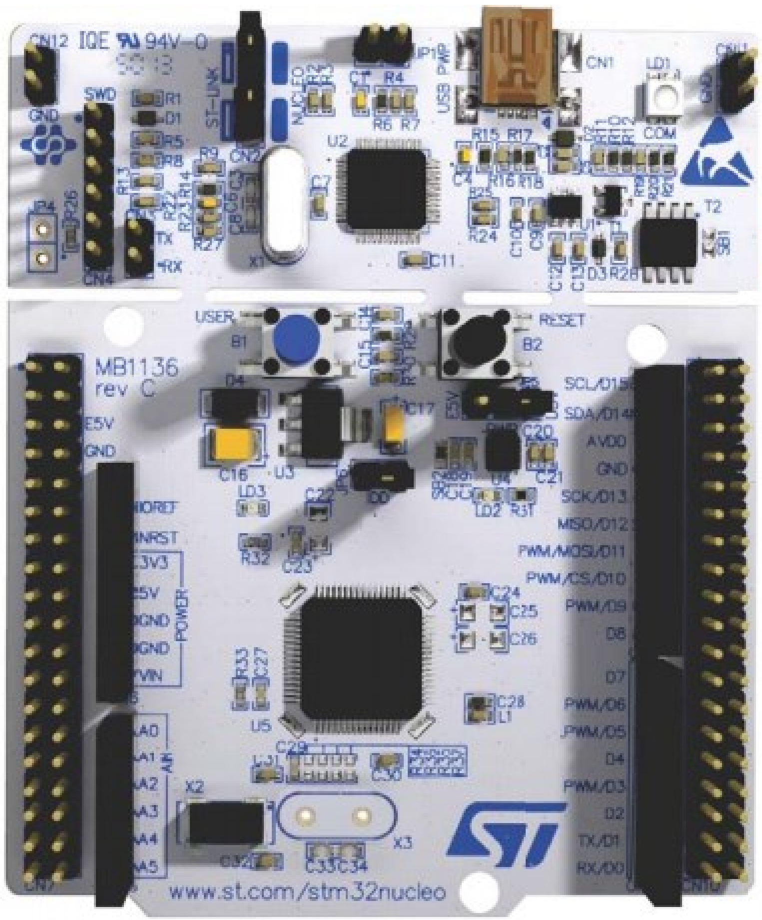
Cortex-M4 CPU:



1. STM32F411xC/E: 256 KBytes / 512KBytes Flash with 128 KBytes SRAM

- Harvard architecture
- Floating Point Unit
  - Single precision floating point unit
- Low power (!)
  - 100μA/MHz

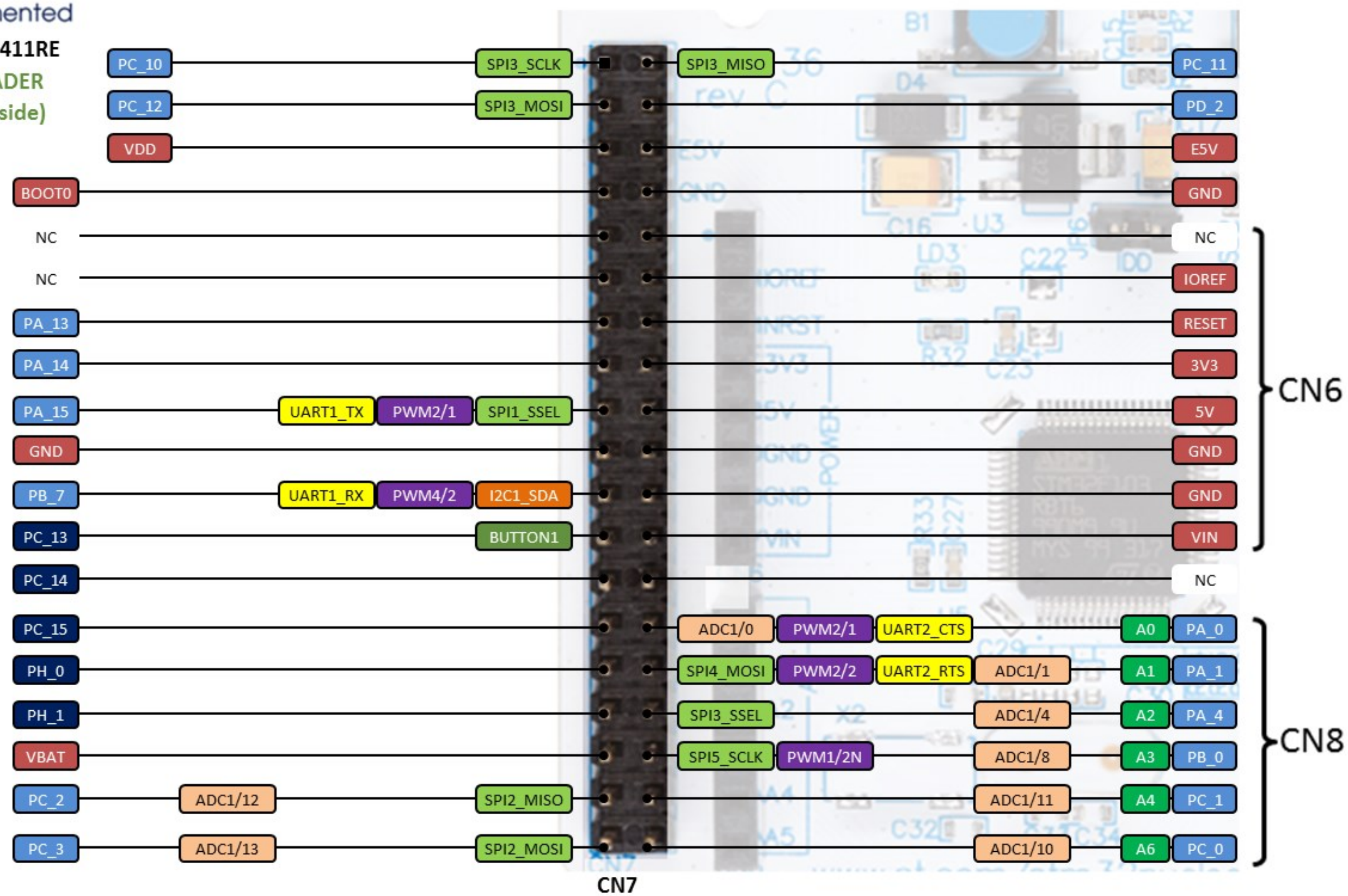
Lab evaluation boards:

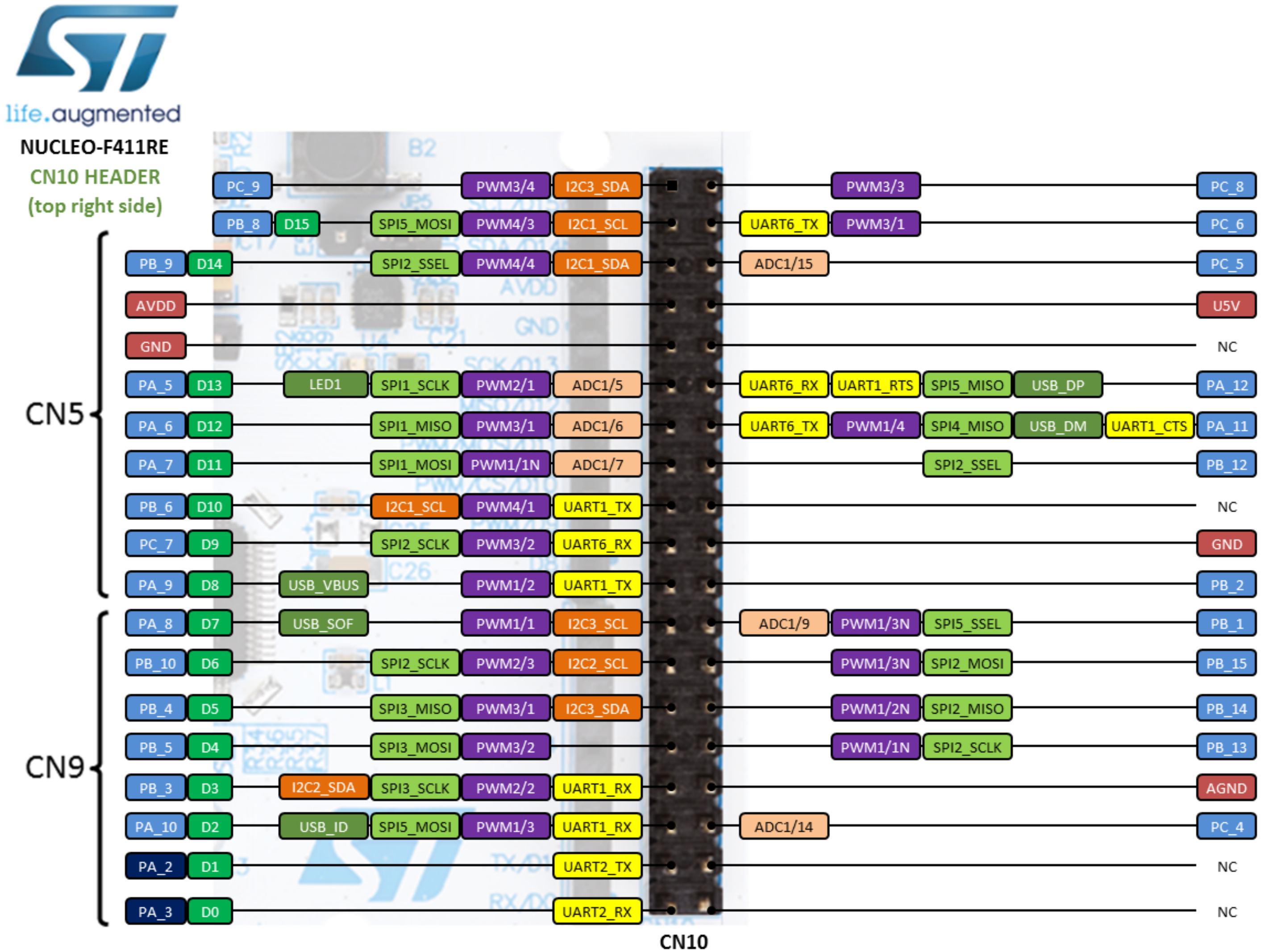


- NUCLEO-F401
- NUCLEO-F411
- NUCLEO-L476

NUCLEO-F411RE

CN7 HEADER
(top left side)

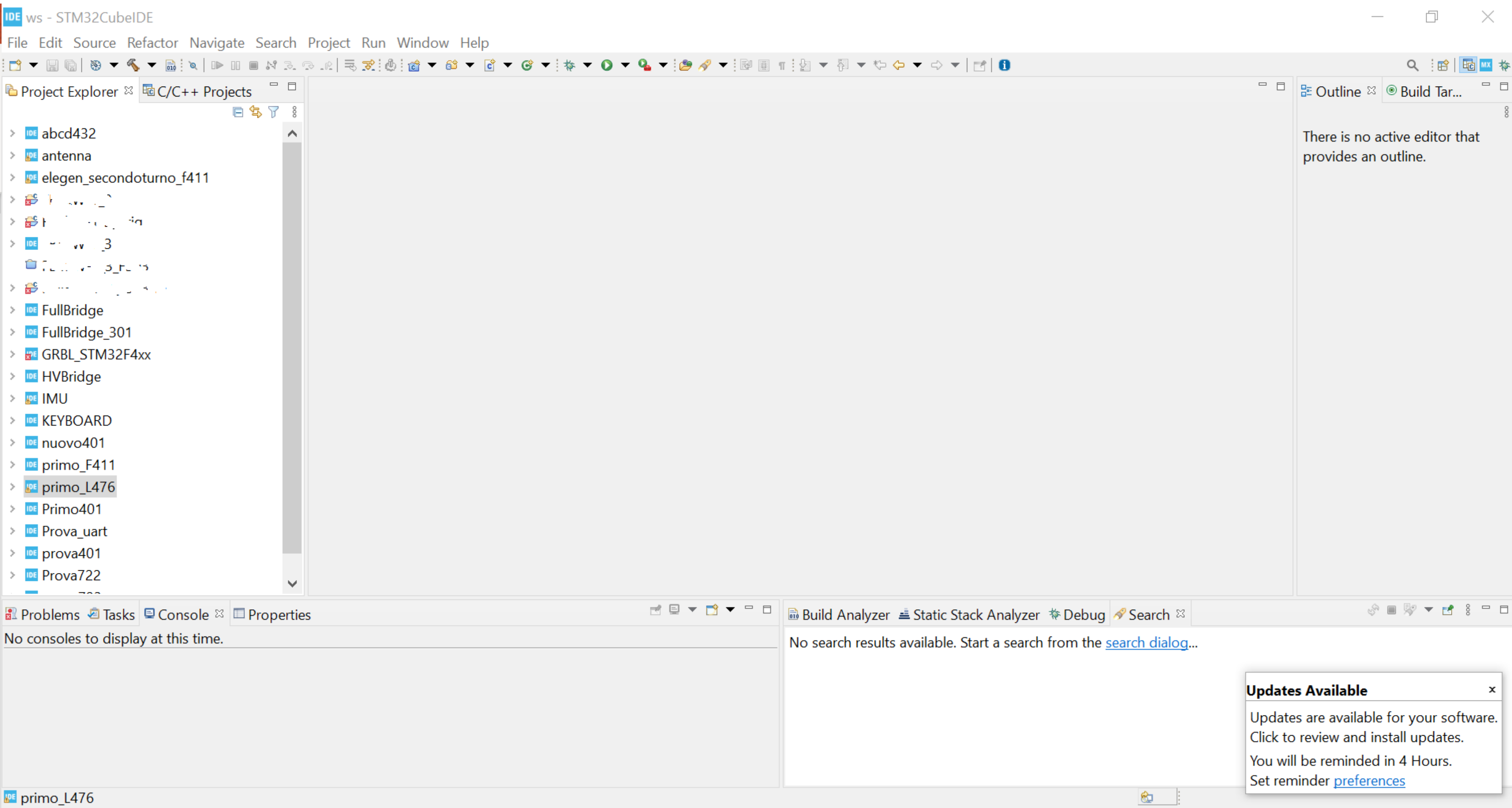



life.augmented
NUCLEO-F411RE
CN10 HEADER
(top right side)
CN5
CN9
CN10
PC_9 PWM3/4 I2C3_SDA | PWM3/3 PC_8
PB_8 D15 SPI5_MOSI PWM4/3 I2C1_SCL | UART6_TX PWM3/1 PC_6
PB_9 D14 SPI2_SSEL PWM4/4 I2C1_SDA | ADC1/15 PC_5
AVDD | U5V
GND | NC
PA_5 D13 LED1 SPI1_SCLK PWM2/1 ADC1/5 | UART6_RX UART1_RTS SPI5_MISO USB_DP PA_12
PA_6 D12 SPI1_MISO PWM3/1 ADC1/6 | UART6_TX PWM1/4 SPI4_MISO USB_DM UART1_CTS PA_11
PA_7 D11 SPI1_MOSI PWM1/1N ADC1/7 | SPI2_SSEL PB_12
PB_6 D10 I2C1_SCL PWM4/1 UART1_TX | NC
PC_7 D9 SPI2_SCLK PWM3/2 UART6_RX | GND
PA_9 D8 USB_VBUS PWM1/2 UART1_TX | PB_2
PA_8 D7 USB_SOF PWM1/1 I2C3_SCL | ADC1/9 PWM1/3N SPI5_SSEL PB_1
PB_10 D6 SPI2_SCLK PWM2/3 I2C2_SCL | PWM1/3N SPI2_MOSI PB_15
PB_4 D5 SPI3_MISO PWM3/1 I2C3_SDA | PWM1/2N SPI2_MISO PB_14
PB_5 D4 SPI3_MOSI PWM3/2 | PWM1/1N SPI2_SCLK PB_13
PB_3 D3 I2C2_SDA SPI3_SCLK PWM2/2 UART1_RX | AGND
PA_10 D2 USB_ID SPI5_MOSI PWM1/3 UART1_RX | ADC1/14 PC_4
PA_2 D1 UART2_TX | NC
PA_3 D0 UART2_RX | NC

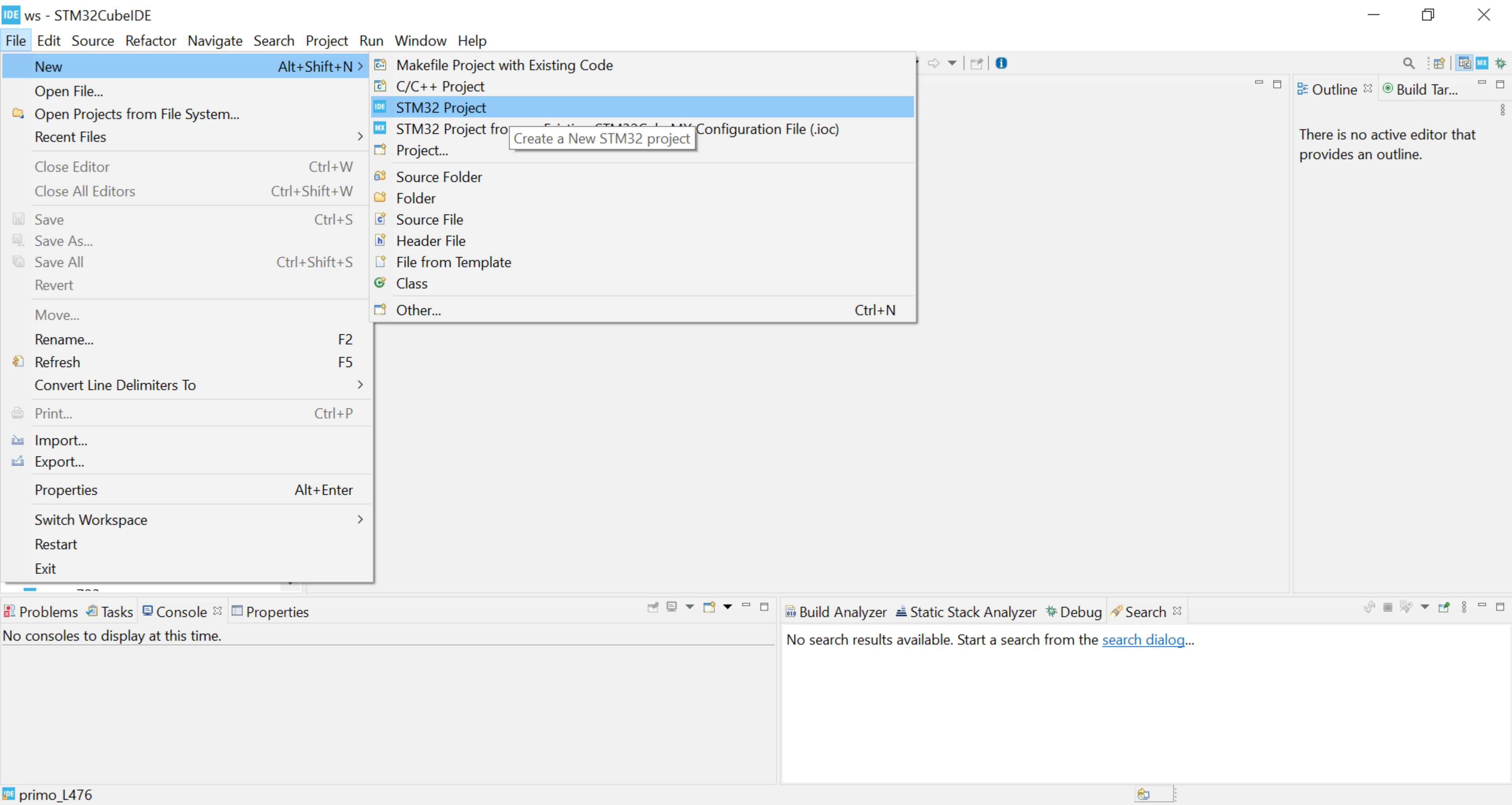
STM32 developement tool:

- STM32CubeIDE

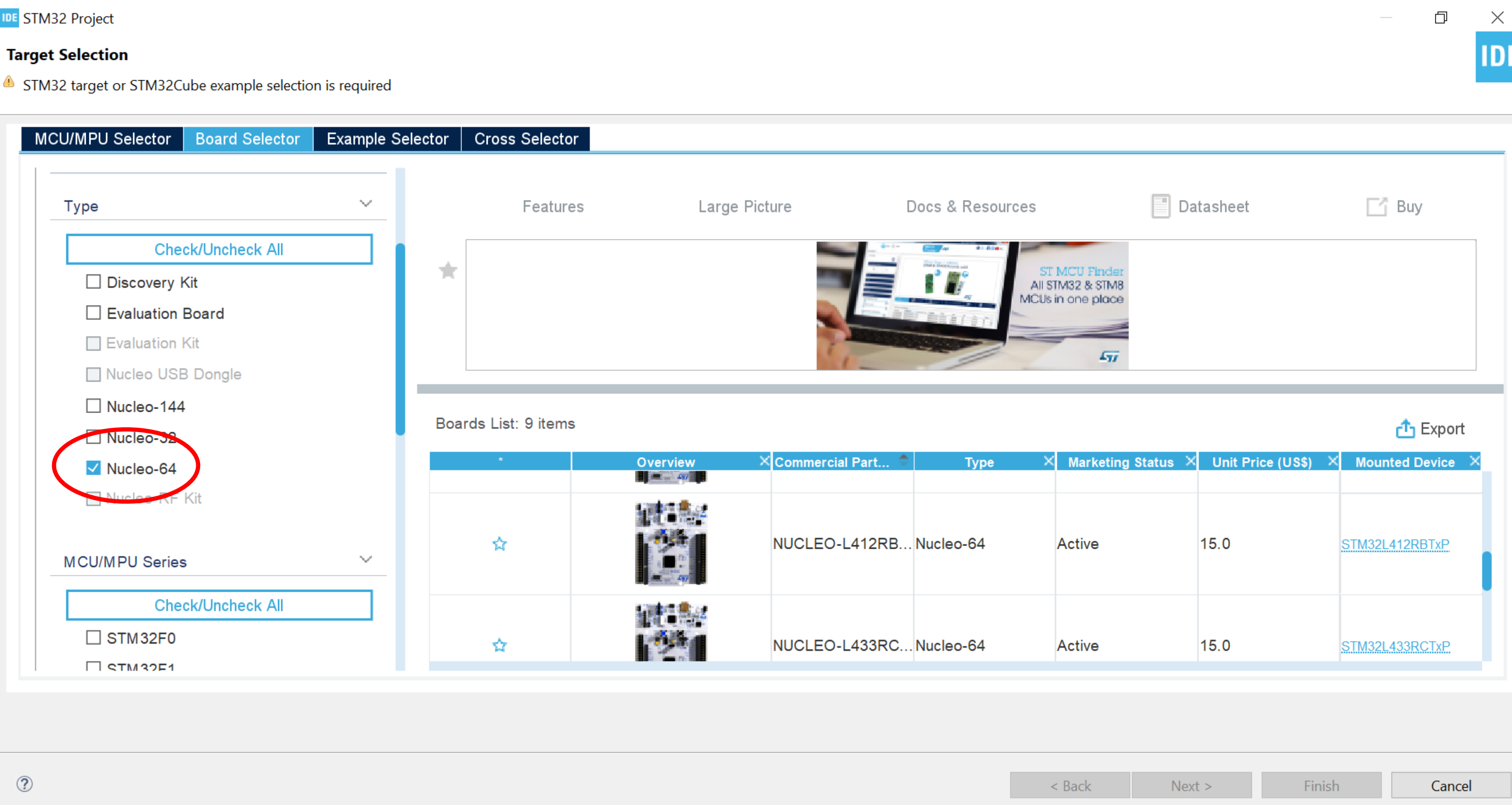
STM32CubeIDE



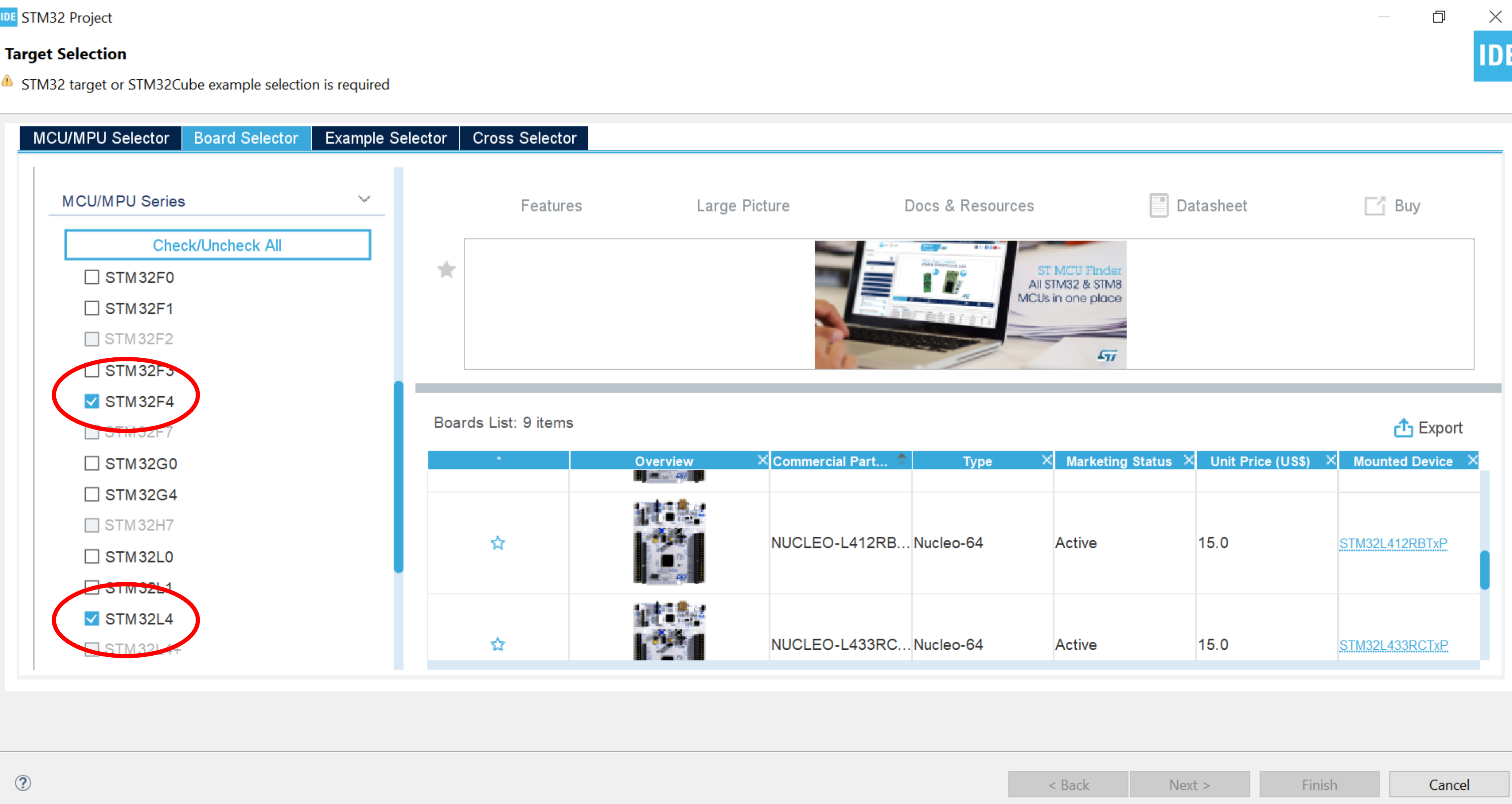
STM32CubeIDE



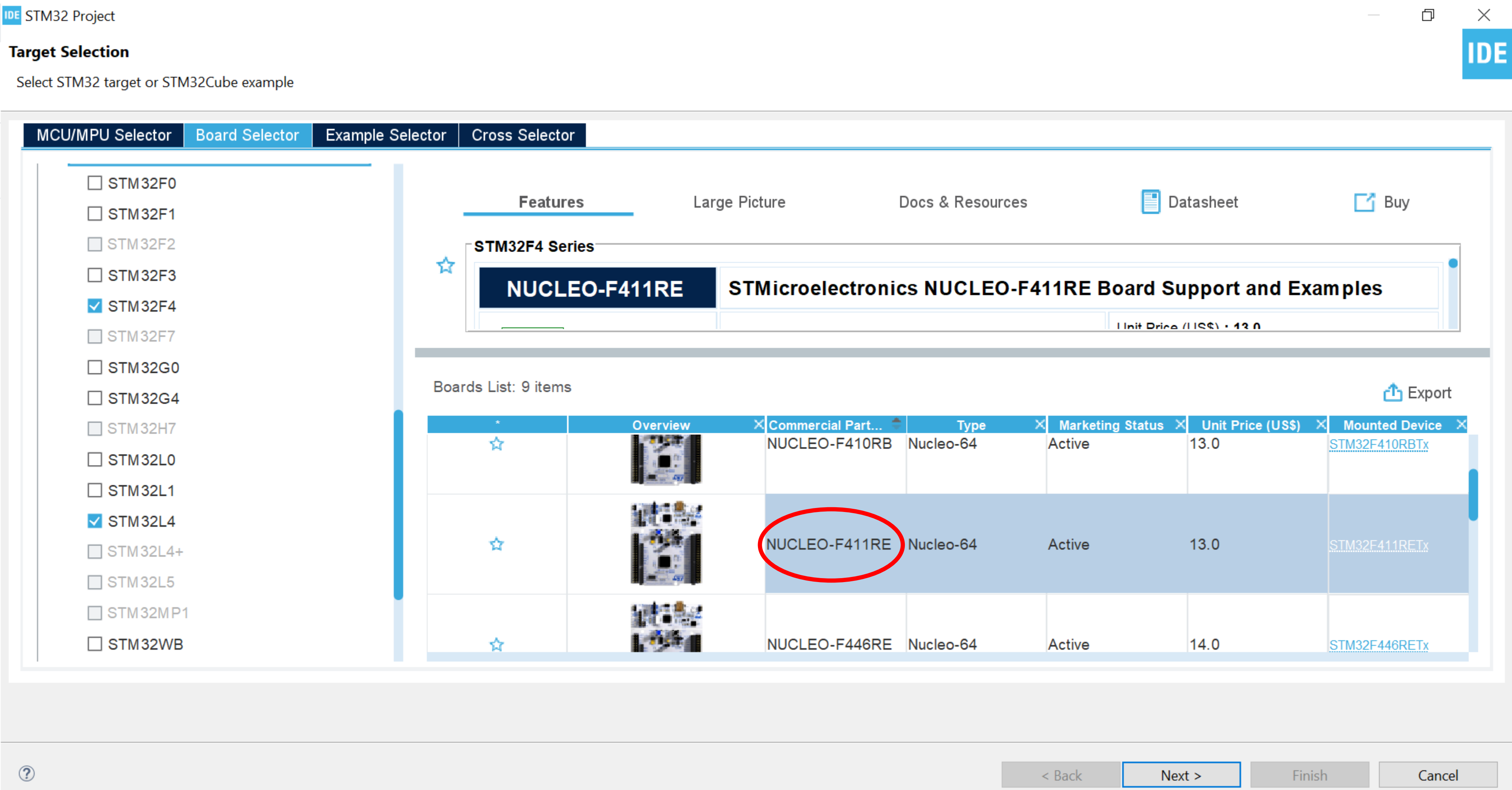
STM32CubeIDE



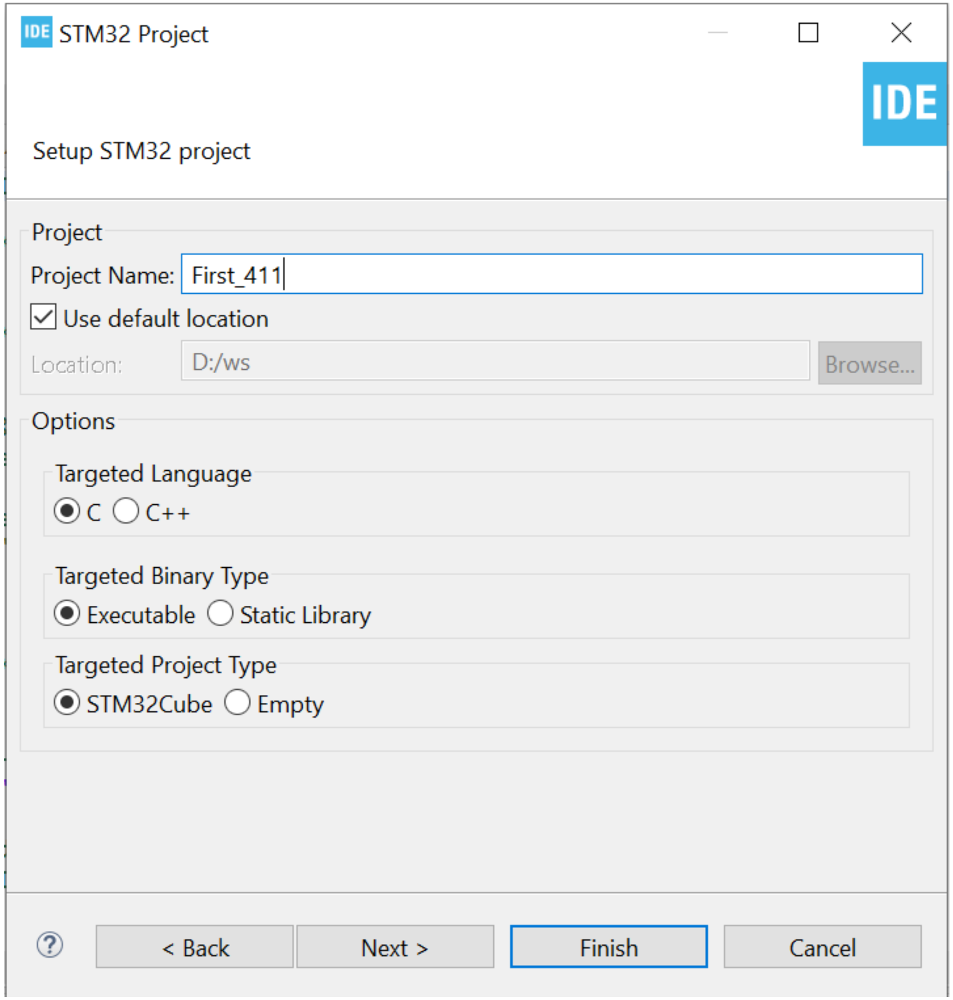
STM32CubeIDE



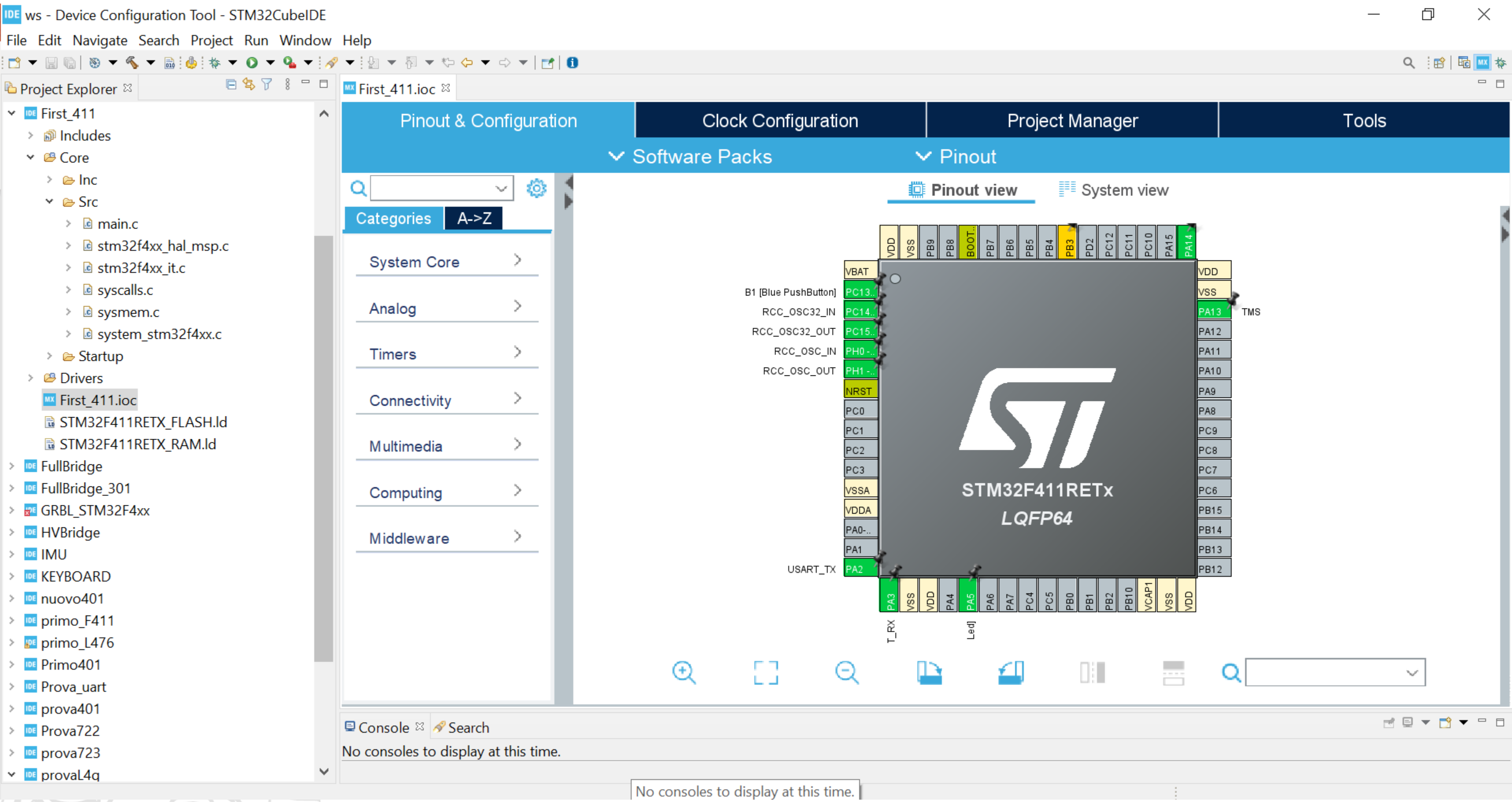
STM32CubeIDE



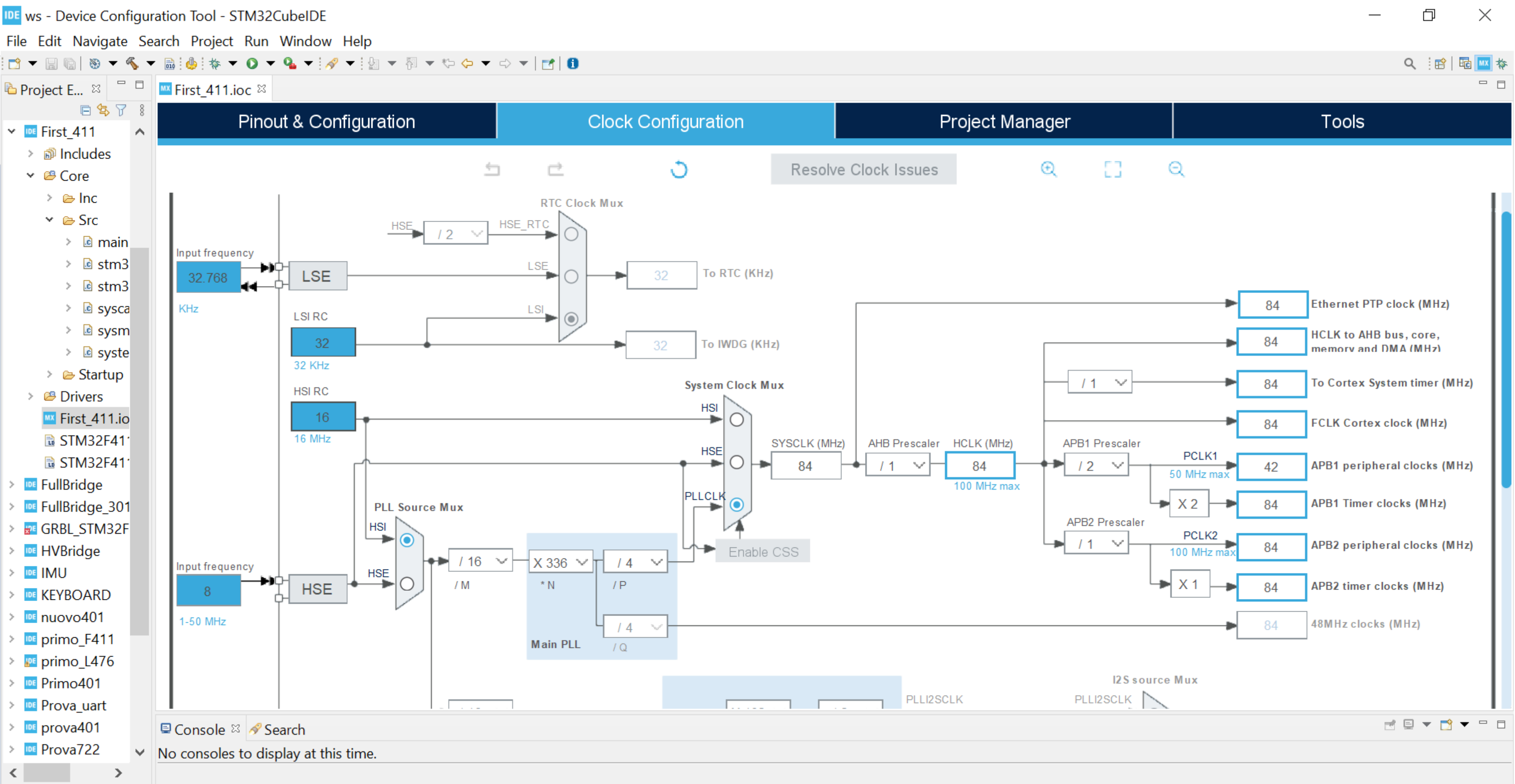
STM32CubeIDE



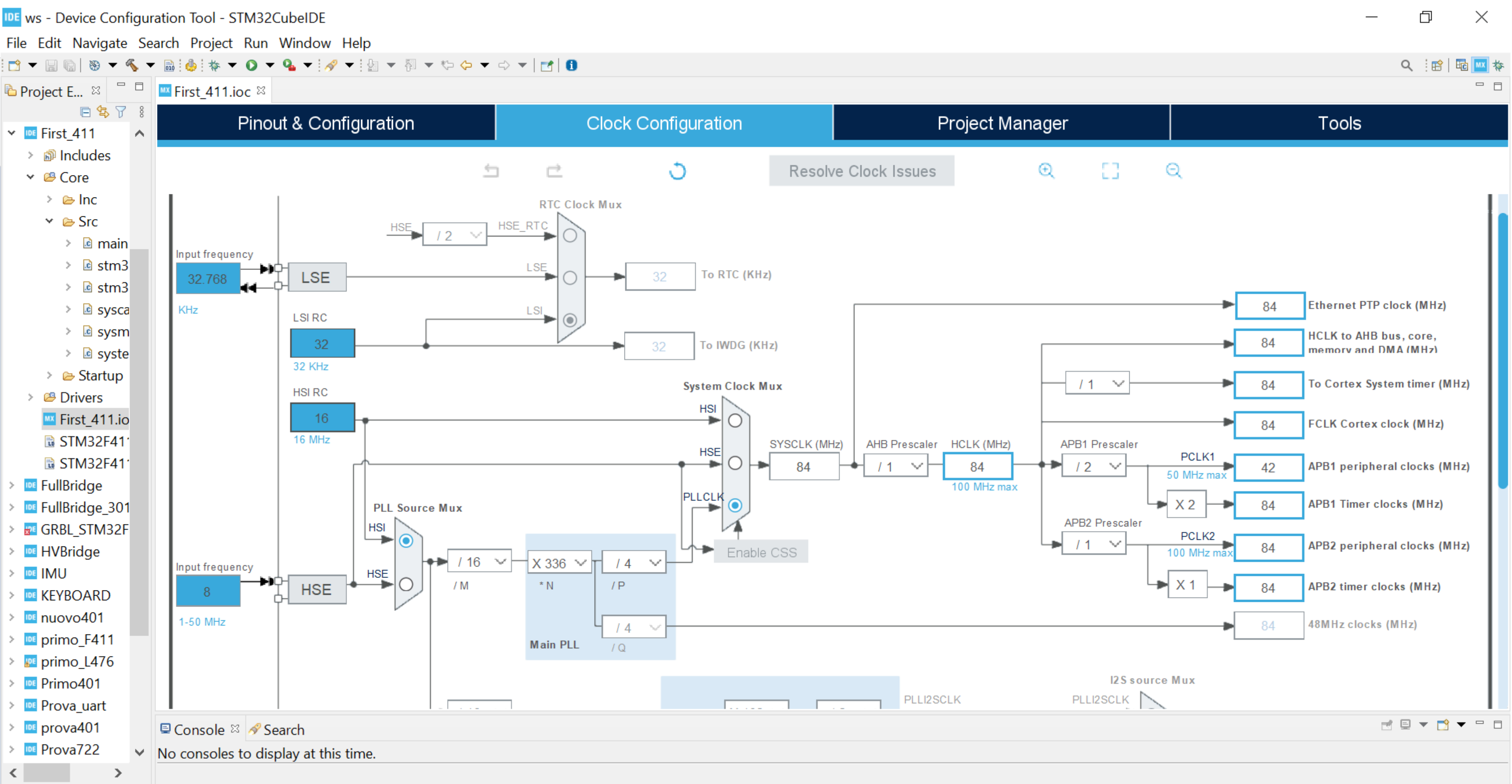
Target Configuration



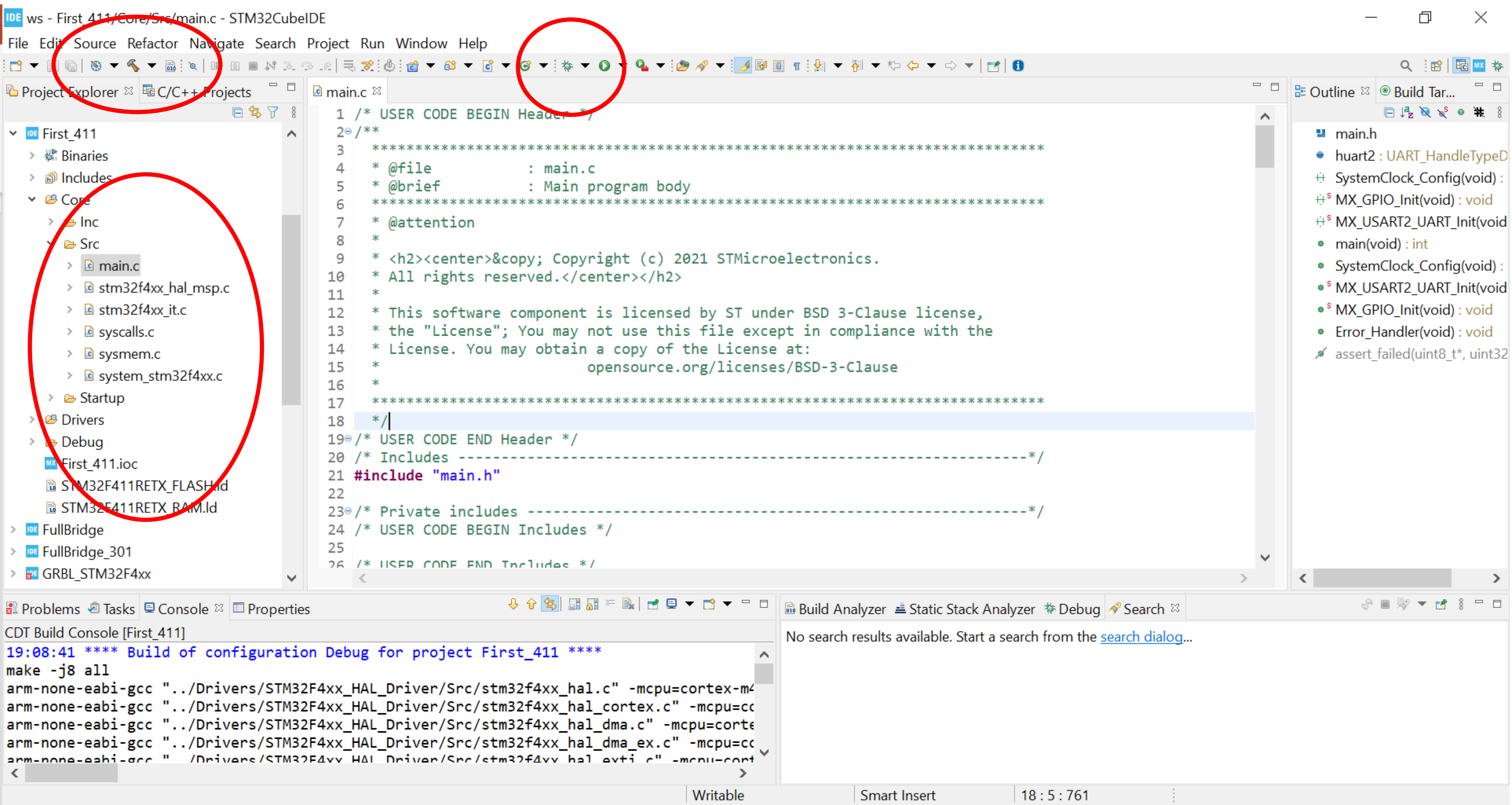
Target Configuration: clock



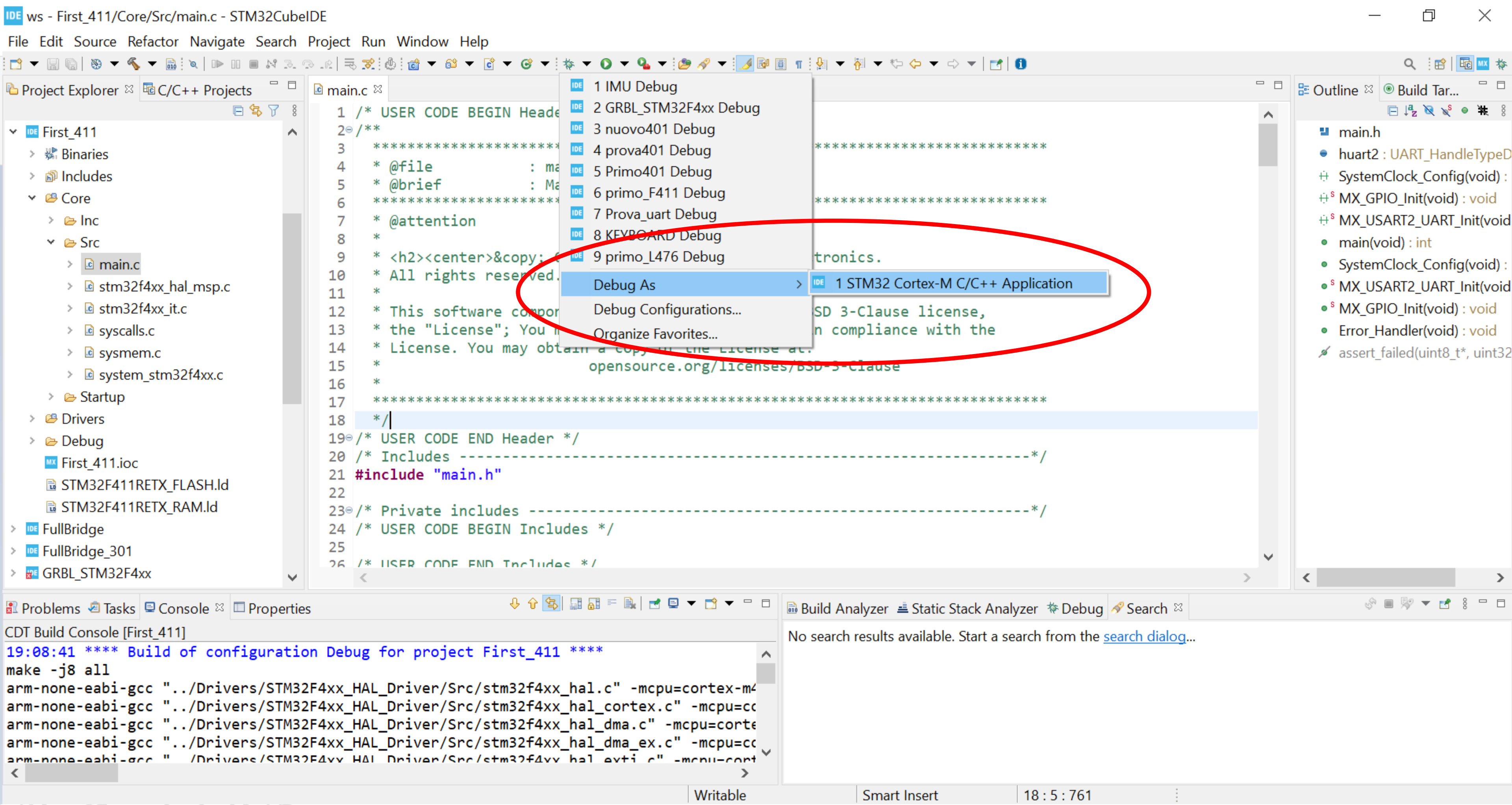
Target Configuration: clock



Development environment



Downloading the code and debugging



Where to write your code…